

FTIR ANALYSIS OF BACTERIAL DNA DATA FOR IDENTIFICATION

Project-2 (EC47004) report submitted

To

Indian Institute of Technology Kharagpur

In partial fulfilment for the award of degree of

Bachelor of Technology

In

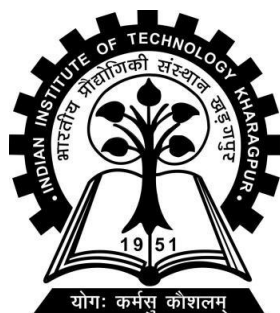
Electronics and Electrical Communication Engineering

By

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Abstract

This report presents an account of my Bachelor Thesis Project (BTP)-1 at Electronics and Electrical Communication Department, Indian Institute of Technology Kharagpur under Prof. Dr. Basudev Lahiri. My main focus in this project is to analyse the bacterial DNA FTIR data and try to find classifiable markers for species and strain identification. Nucleotide Composition FTIR Data analysis gave us a way to find DNA markers in FTIR Data.

A series of statistical tests were conducted which revealed clean and clustered data points in bacterial DNA, whose Principal Components were used as features for classification with a reasonably high accuracy (89%) and further trained a unified neural network that can classify into species and strain type with an accuracy of 96%. They also revealed a very complex statistic relationship.

Assuming a linear relationship between FTIR intensity and concentration, NNLS Regression was done to solve the optimization of a reconstructed plot from weight average of poly spectra to be the best fit for the mutated Data to obtain a potential DNA marker in FTIR data.

Shift magnitude, the amount of red or blue shift in a certain sample's FTIR as compared to the mean of all poly spectra was employed as a novel way for DNA nucleotide identification through FTIR Data.

These findings were then used to check which nucleotides present themselves in these vibrational shifts in the bacterial data.

The code for analysis on [bacterial data](#) and [Nucleotide composition data](#) are hosted on Github.

Acknowledgement

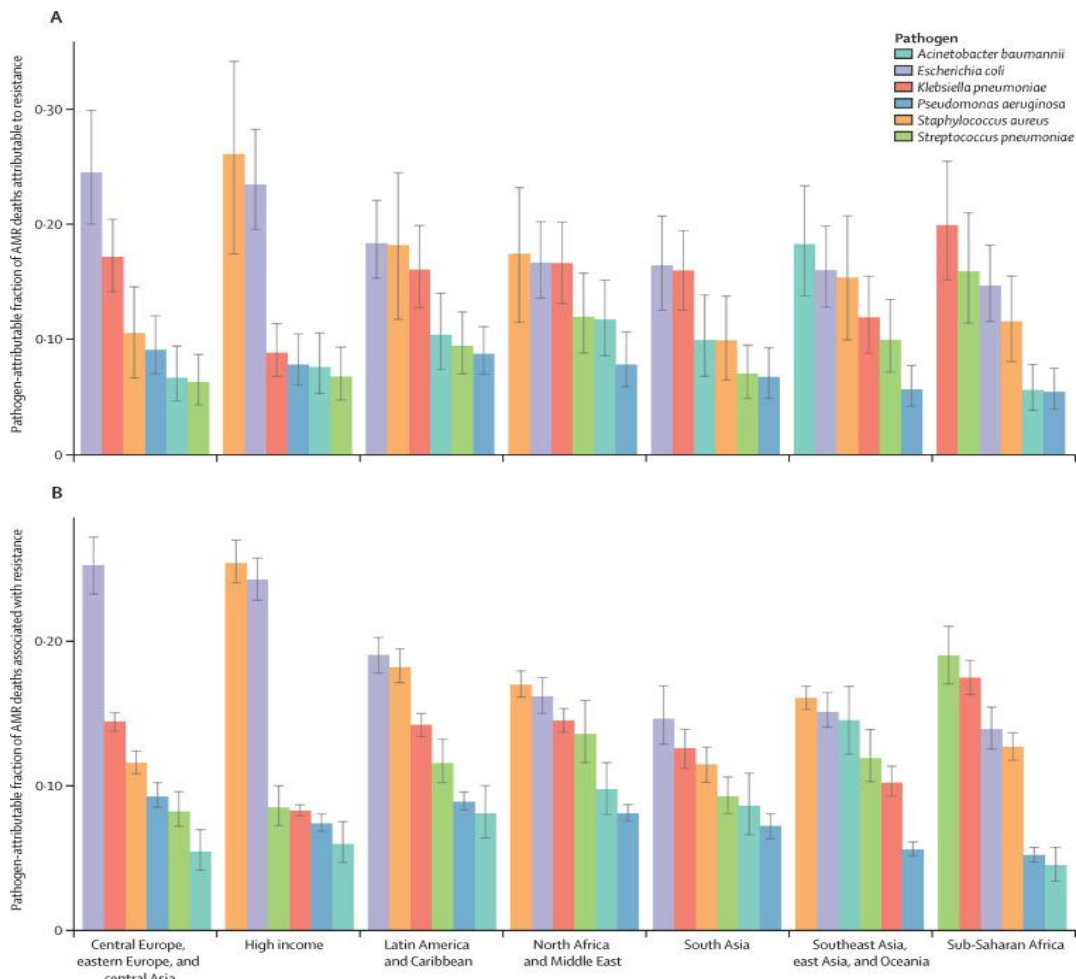
Foremost, I am incredibly grateful to Prof. Basudev Lahiri (Faculty Advisor and Project Guide), Prof. Tarun Kanti Bhattacharya (HoD, E&ECE) and Prof. Arun Chakraborty (Dean of Student Wellbeing) for providing me this project at E&ECE department of IIT Kharagpur and all the professors who gave me time and resources. It is only because of their express support and encouragement, have I been able to do this internship successfully.

My sincere thanks to my seniors and mentors Subhanita Roy and Souvik Das for providing me with this meticulous data and for guiding me through difficult parts of the internship by thoughtful suggestions, insightful comments and corrections. I appreciate their helpful attitude towards me in my time of need. I also want to thank Shroyon Das Gupta, a past student who had helped Souvik Das get the experimental gene composition data.

I also thank everyone from the Institute fraternity who stepped up for me as their family.

Introduction

Antibiotics were the miracle drugs that gave a chance to fight against a range of bacterial infections. But mutations occur all the time and more frequently in bacterial colonies. Antimicrobial resistance (AMR) was associated with an estimated **4.95 million deaths globally in 2019**, including **1.27 million deaths directly attributable** to bacterial AMR, making it one of the leading global public health threats. (Murray, 2022)



Keeping track of these mutations is essential for discovering new drugs to combat these drug-resistant pathogens. Whole-genome sequencing (WGS), while highly accurate, generally requires **7–9 hours to 2–5 days** depending on sequencing platform and downstream bioinformatic analysis. In addition, sequencing instruments typically cost **US\$3,000–1,000,000**, while per-sample sequencing costs are approximately **US\$50–238**, making routine large-scale implementation resource-intensive. (Afonso, 2025)

Fourier Transform-Infrared (FTIR) produces spectra that are distinct for different species of bacteria. Now the question is whether there is enough variation to provide markers for classification in the spectral data.

Lit Review

I focused on papers that deal with FTIR for species and strain type classification and base pair identification.

(Naumann, 2001) had given the wavenumber regions that have the most relation to base pair vibrations.

(L. Mariey, 2001) had used HCA and PCA to assess similarity across the strain types and species. (Mareike Wenning, 2013) had used machine learning techniques to find out applications and limitations in FTIR technology for the identification of pathogens.

(Amin Kassem, 2023) provided me with the wavenumbers that correspond to specific vibrations in base pairs. Most of the approaches I saw had used peak shifts as a feature.

Possible assignment of some bands frequently found in microbial IR spectra (peak frequencies have been obtained from the second derivative spectra). (Amin Kassem, 2023)

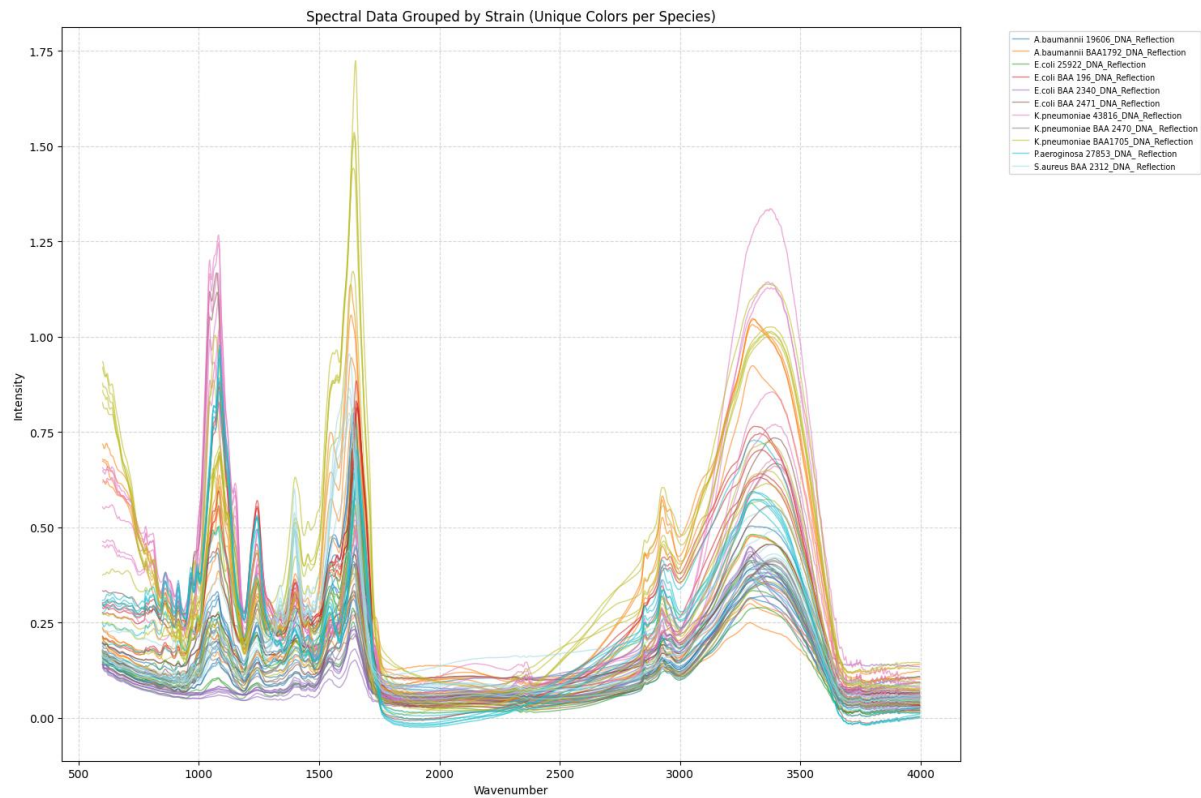
Assignment	Frequency (cm ⁻¹)
C-H stretching peaks	
Alkane C-H stretching	2,850–2,960
Alkene C-H stretching	3,010–3,100
Alkyne C-H stretching	3,300
O-H and N-H stretching peaks	
Alcohols and phenols O-H stretching	3,200–3,600
Carboxylic acids O-H stretching	2,500–3,300
Amines N-H stretching	3,300–3,500
C=O stretching peaks	
Ketones and aldehydes C=O stretching	1,680–1,750
Carboxylic acids C=O stretching	1,700–1,750
Amides C=O stretching	1,630–1,690
C-N stretching peaks	
Aromatic C-N stretching	1,300–1,350
Aliphatic C-N stretching	1,000–1,300
C=C stretching peaks	
Alkene C=C stretching	1,620–1,680
Aromatic C=C stretching	1,450–1,600
N-H bending peaks	
Amines N-H bending	1,560–1,640
Amides N-H bending	1,550–1,670

Assignment	Frequency (cm ⁻¹)
C-H bending peaks	
Alkanes C-H bending	1,370–1,470
Alkenes C-H bending	960–1,200
O-H bending peaks	
Alcohols and phenols O-H bending	1,350–1,450
Fingerprint region peaks	700–900

Data

Bacterial DNA Data

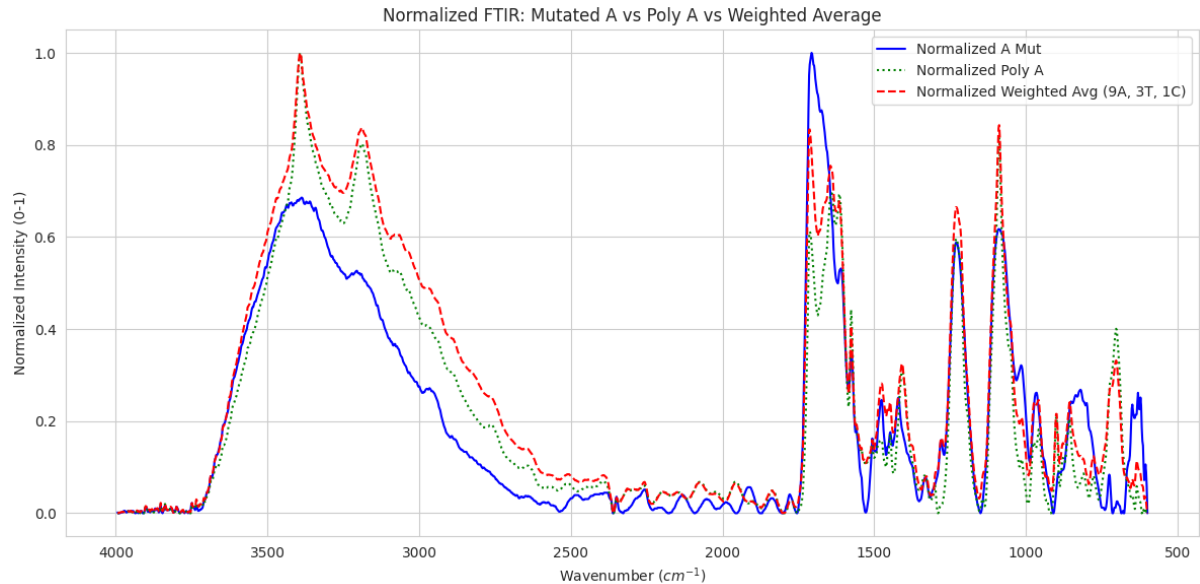
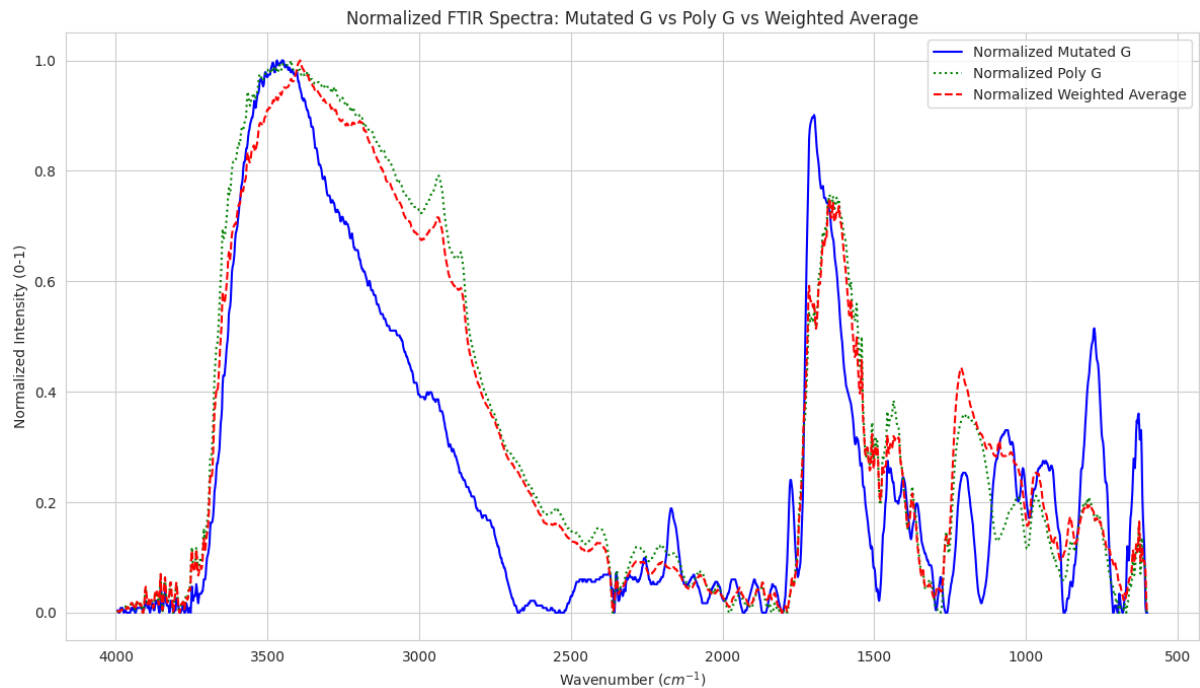
I had received the following FTIR spectra of different Bacterial DNA samples, some of which are drug sensitive and some drug resistant, from Subhanita Roy.

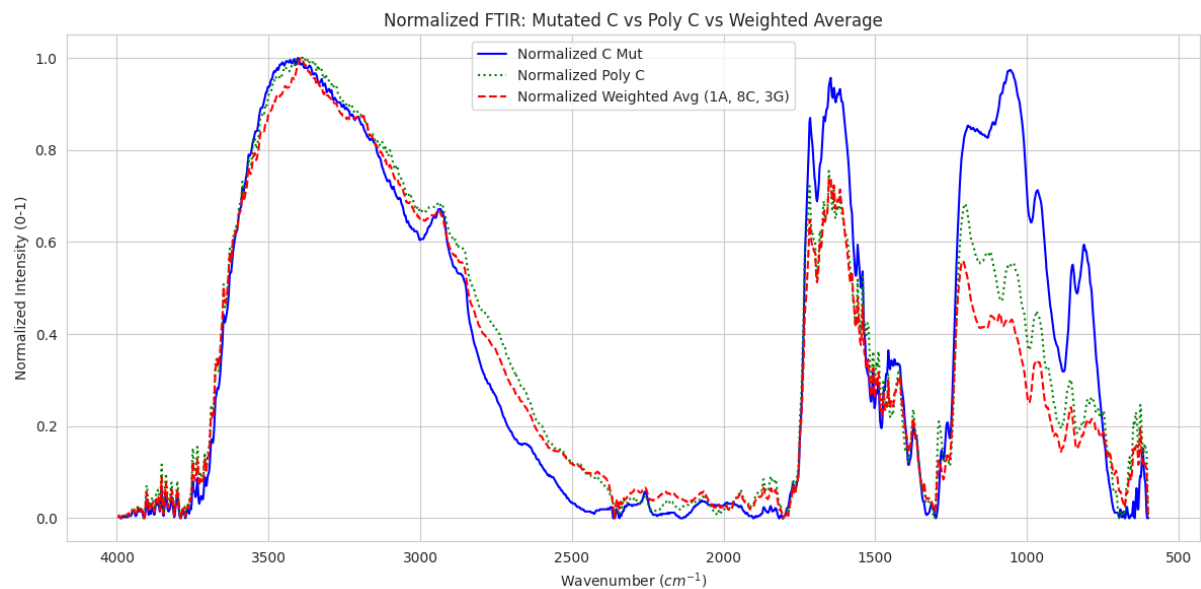
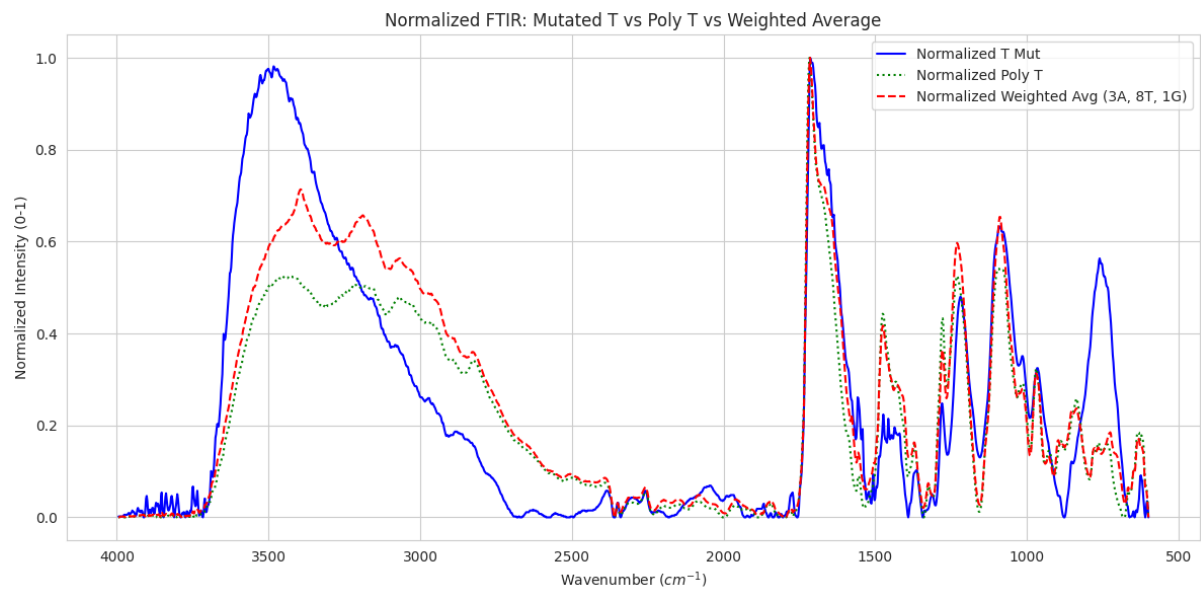


My **aim** here is to find out which components of the spectra can classify the data into drug sensitivity types and species types.

DNA Composition Effect on FTIR Data

I received FTIR spectral data from samples of artificially constructed DNA strands from Souvik Das, some consisting purely of a single base pair, others with mutated strands containing a mix of 3 base pairs, and a weighted average and median of the pure spectral data for each mutated sample.



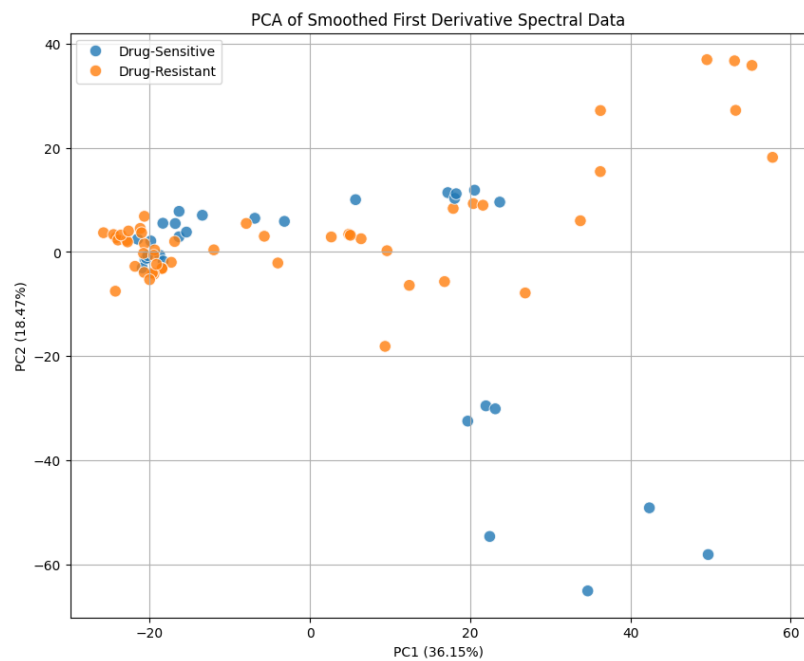
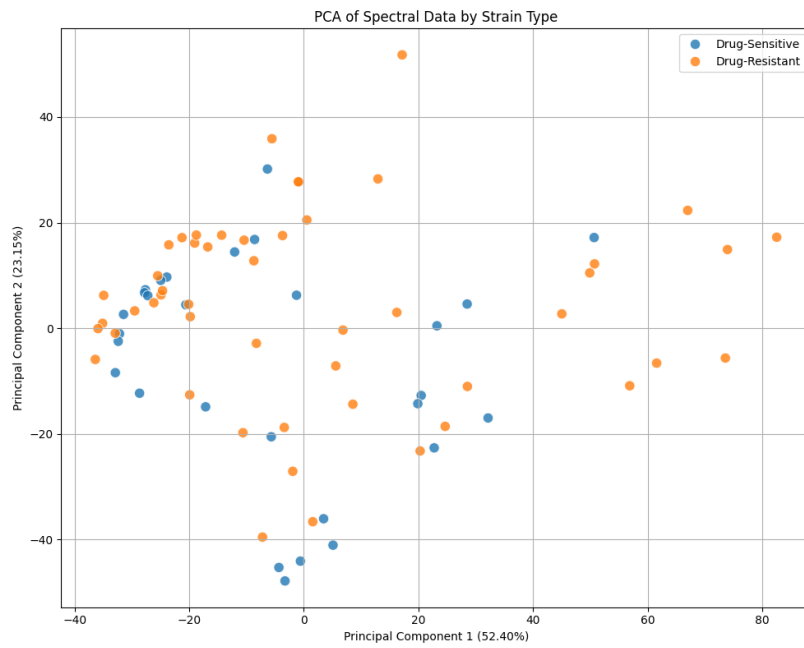


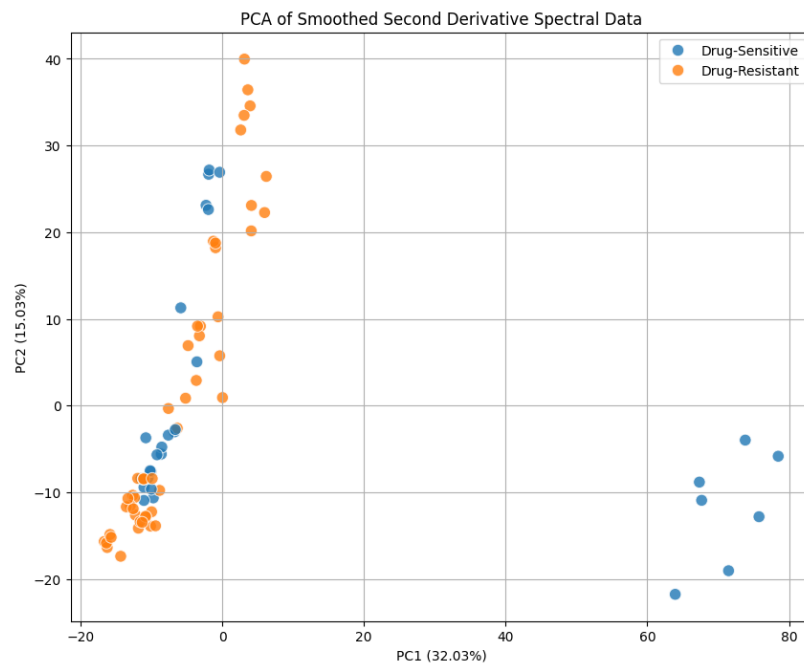
Aim here is to find a way be able to confidently know the nucleotide composition of the DNA sample through FTIR Data and to use these results to improve upon the analysis done on the Bacterial Data.

Statistic analysis

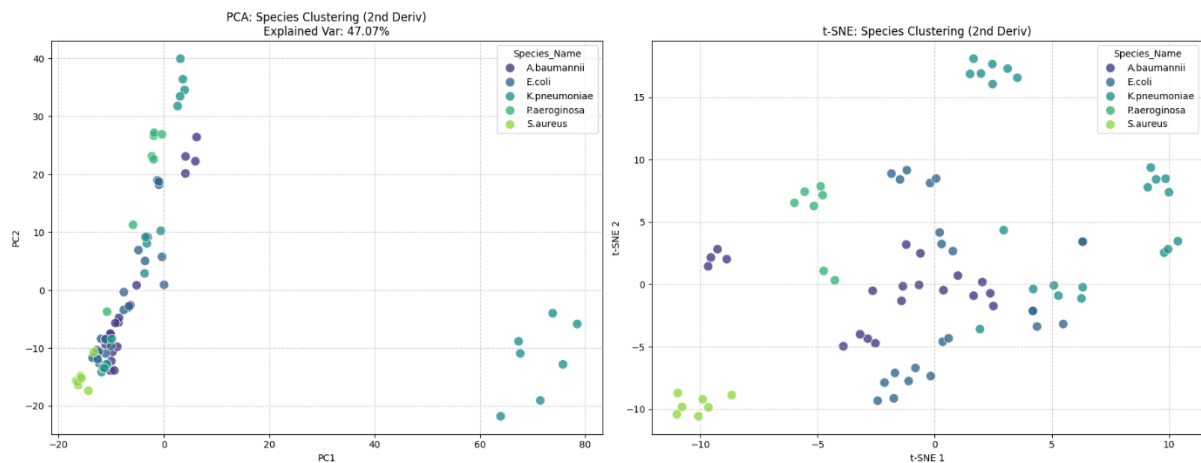
Bacterial Data

I wanted to know if there are any statistical variations in the spectral Data that can provide meaningful classification of the strain types, so I conducted a Principal Component Analysis on the original data and its first and second derivatives.

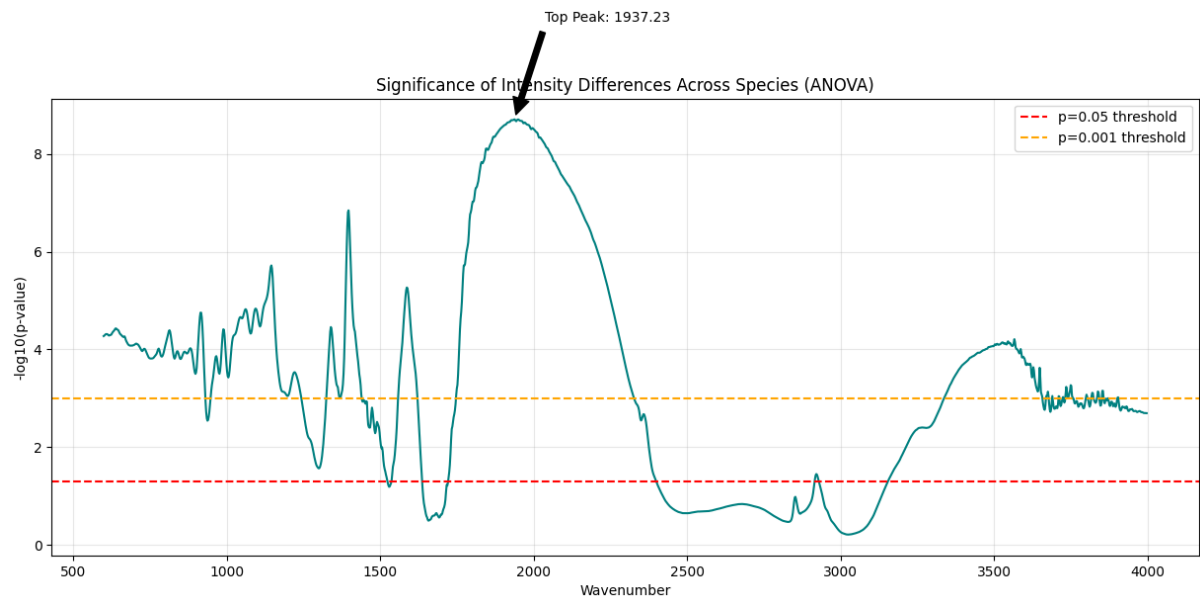




I also did the same for species variation and as you can see very neat clusters.

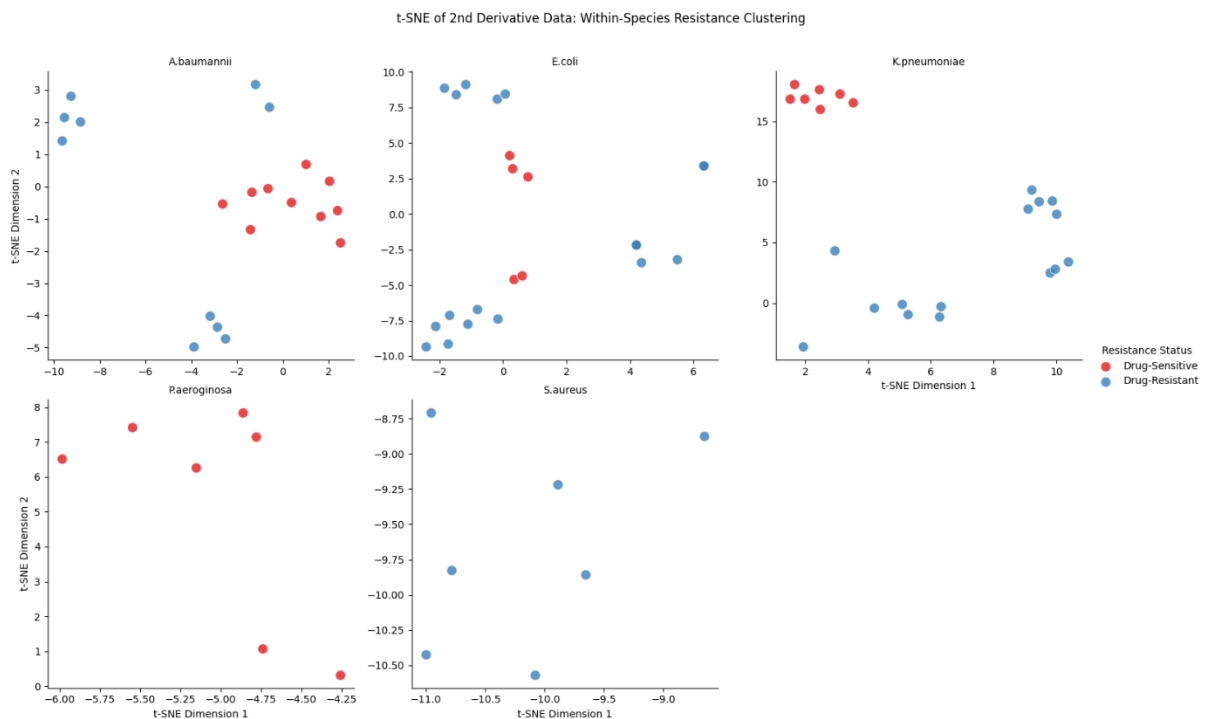


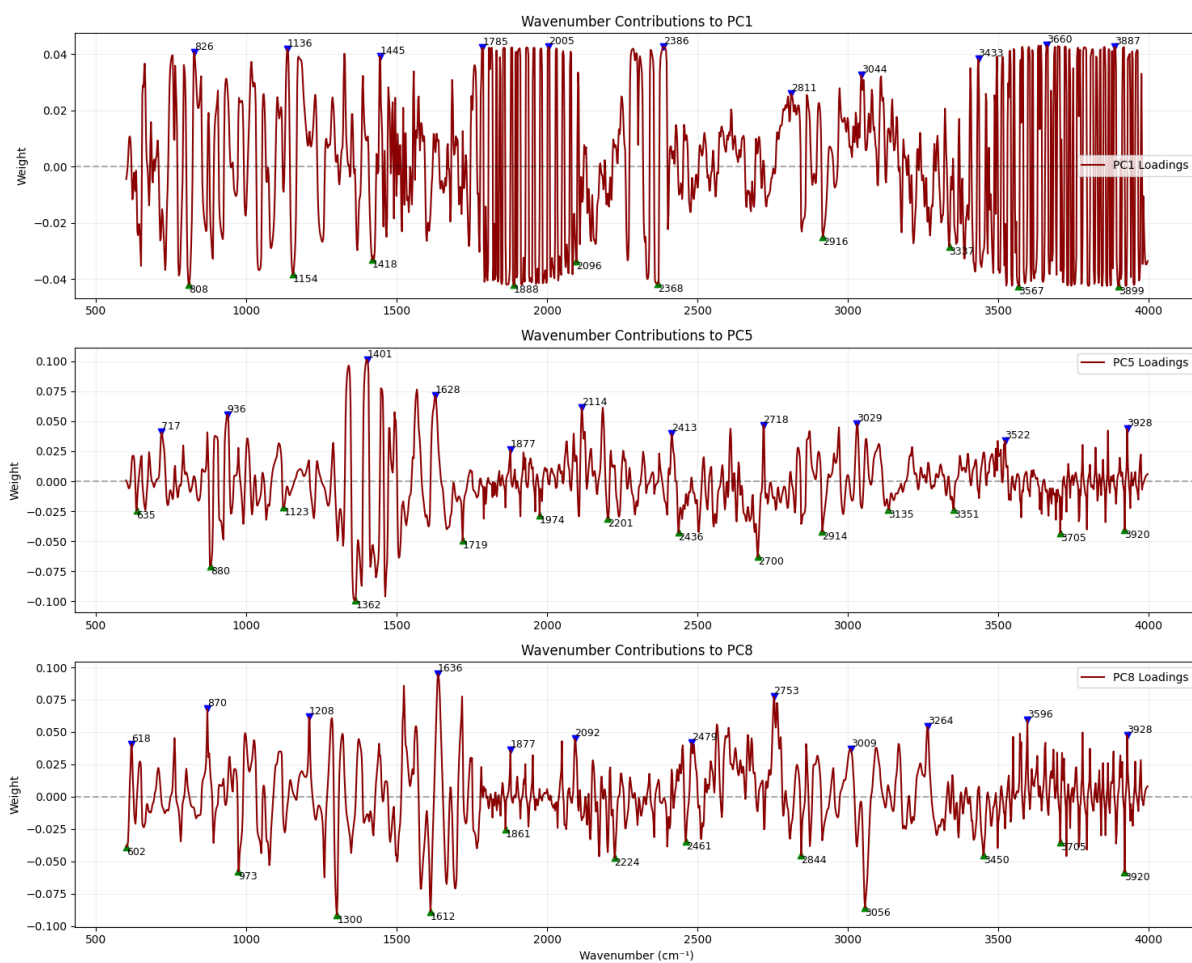
I also had conducted an ANOVA test to find out where most of this variation is concentrated in which regions of the spectra can we find most variation that is not just an artifact.



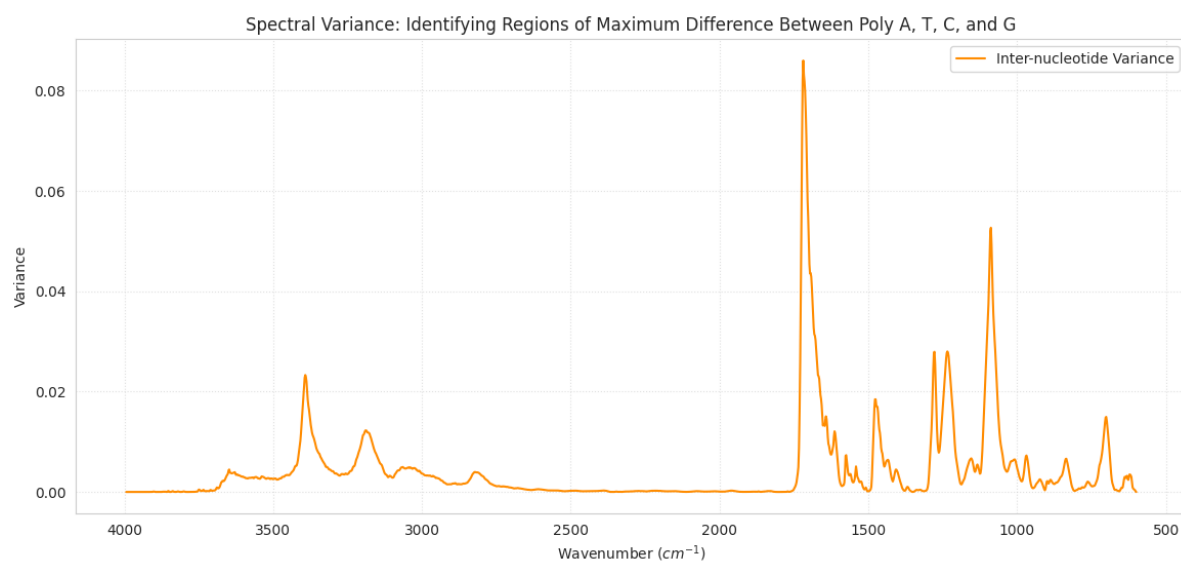
As we can see, most variation comes from regions between 1000 cm^{-1} to 2300 cm^{-1} . Sharp peaks in the plot above indicate that a single or a set of very close wavenumbers have statistic information to be exploited.

I also tried to see if there is any clustering in the t-SNE in the second derivative of the original data. We can observe very clean clusters, which tells us that there are components in the data that can classify this data efficiently.



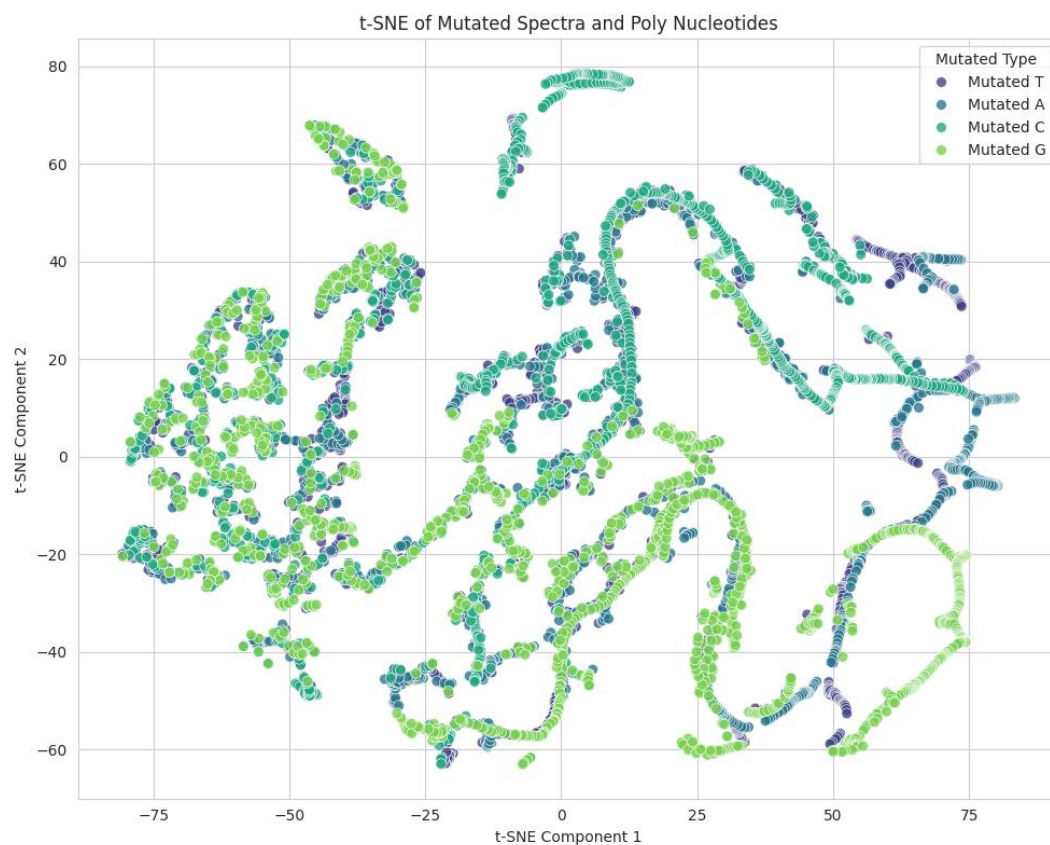


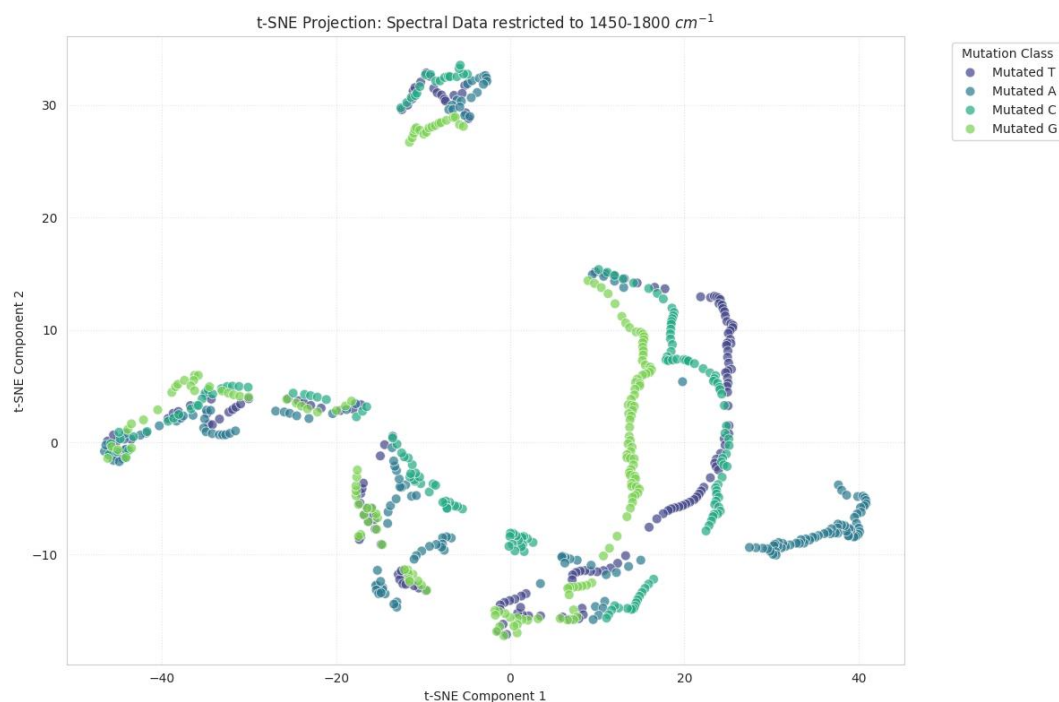
I also analysed the wave number contributions to Principal Components, where the region of 1000 to 2000 cm^{-1} appears to contribute much to PCs. The plot also shows specific wavenumbers to look out for in such a classification.



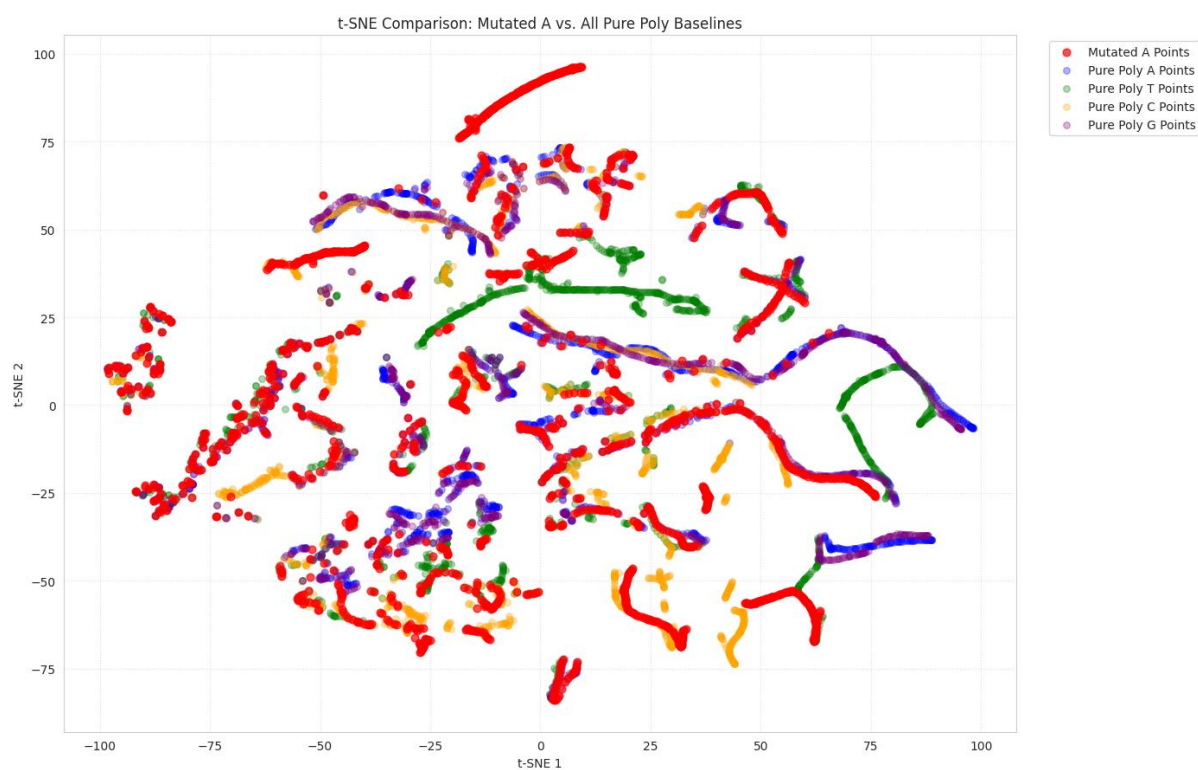
Nucleotide Composition Data

I had conducted a t-SNE of mutated and pure poly spectral data to see if there are any clusters that can help in classification of mutated types and in predicting the composition of base pairs through FTIR. I also produced one with relevant wavenumber for base pair identification, between 1450-1800 cm^{-1} . The **1450–1800 cm^{-1}** region corresponds primarily to nucleic acid and protein vibrational modes, including vibrations associated with nucleobases and amide groups, making it highly informative for bacterial and DNA characterization using FTIR spectroscopy. (Naumann, 2001)





t-SNE of these data shows unusual connected clustering and considerable levels of overlap. So I wished to see if any points of pure samples overlap with mutated ones. It shows a complex relation in the data. As we can see, most of the pure G points do not overlap with the mutated A points, but mutated A overlaps with some poly A, C, and T clusters, which actually compose the mutated gene.



Methods and Performance

Bacterial DNA

I had used a regression model with PCA components as features. Due to high statistical variability in clusters, regression gives very good results. I tried to classify on strains and species.

Detailed Classification Report:

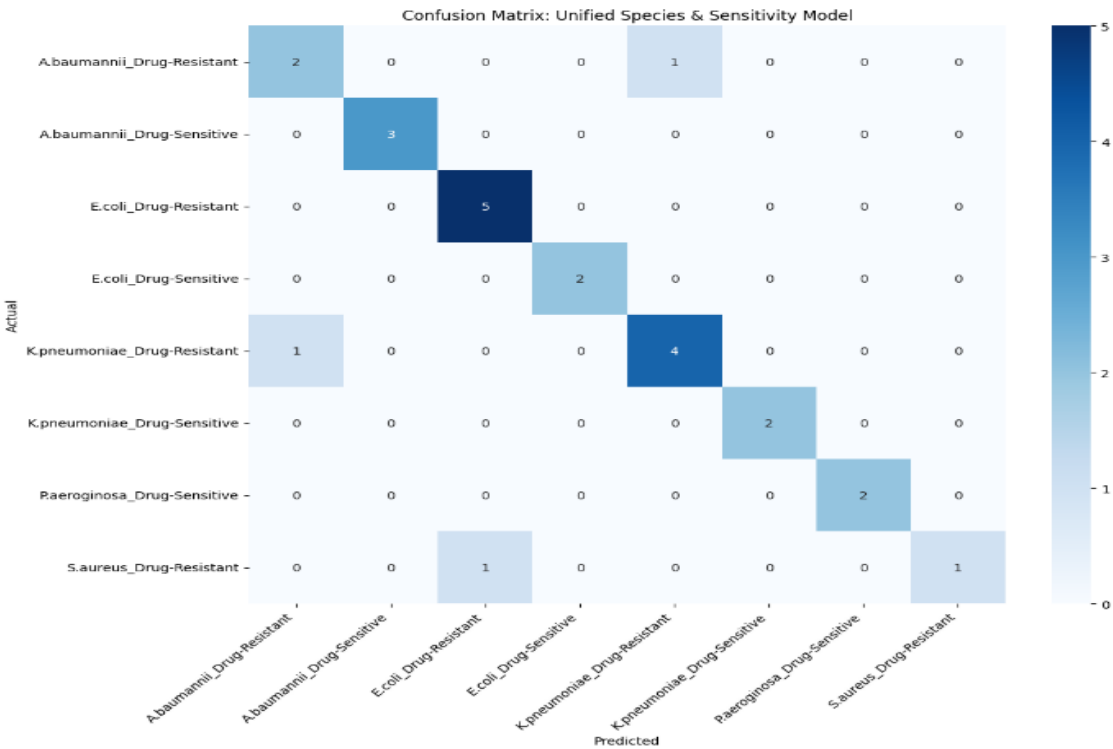
	precision	recall	f1-score	support
A.baumannii	1.00	0.83	0.91	6
E.coli	0.88	1.00	0.93	7
K.pneumoniae	1.00	1.00	1.00	7
P.aeruginosa	1.00	1.00	1.00	2
S.aureus	1.00	1.00	1.00	2
accuracy			0.96	24
macro avg	0.97	0.97	0.97	24
weighted avg	0.96	0.96	0.96	24

Classification Report:second derivative

	precision	recall	f1-score	support
Drug-Resistant	0.93	1.00	0.96	25
Drug-Sensitive	1.00	0.87	0.93	15

accuracy			0.95	40
macro avg	0.96	0.93	0.95	40
weighted avg	0.95	0.95	0.95	40

Unified Model Accuracy: 0.8750



I had wished to increase the accuracy of the model on species and strains. So, I tried training a random forest model and neural network model and tuned it for better results.

Tuned Random Forest Classification Report:				
	precision	recall	f1-score	support
Drug-Resistant	0.69	0.85	0.76	13
Drug-Sensitive	0.75	0.55	0.63	11
accuracy			0.71	24
macro avg	0.72	0.70	0.70	24
weighted avg	0.72	0.71	0.70	24

Classification Report for Neural Network (on validation set):				
	precision	recall	f1-score	support
Drug-Resistant	0.82	0.47	0.60	38
Drug-Sensitive	0.39	0.76	0.52	17
accuracy			0.56	55
macro avg	0.61	0.62	0.56	55
weighted avg	0.69	0.56	0.58	55

These work worse than the baseline regression model, which is understandable given the number of samples available for training. The neural network performed as well as a random guess.

I tried to tune the neural network further until I received 96% accuracy on unified strain and species classification. The details and confusion matrix are on the next page.

Training Unified Neural Network...

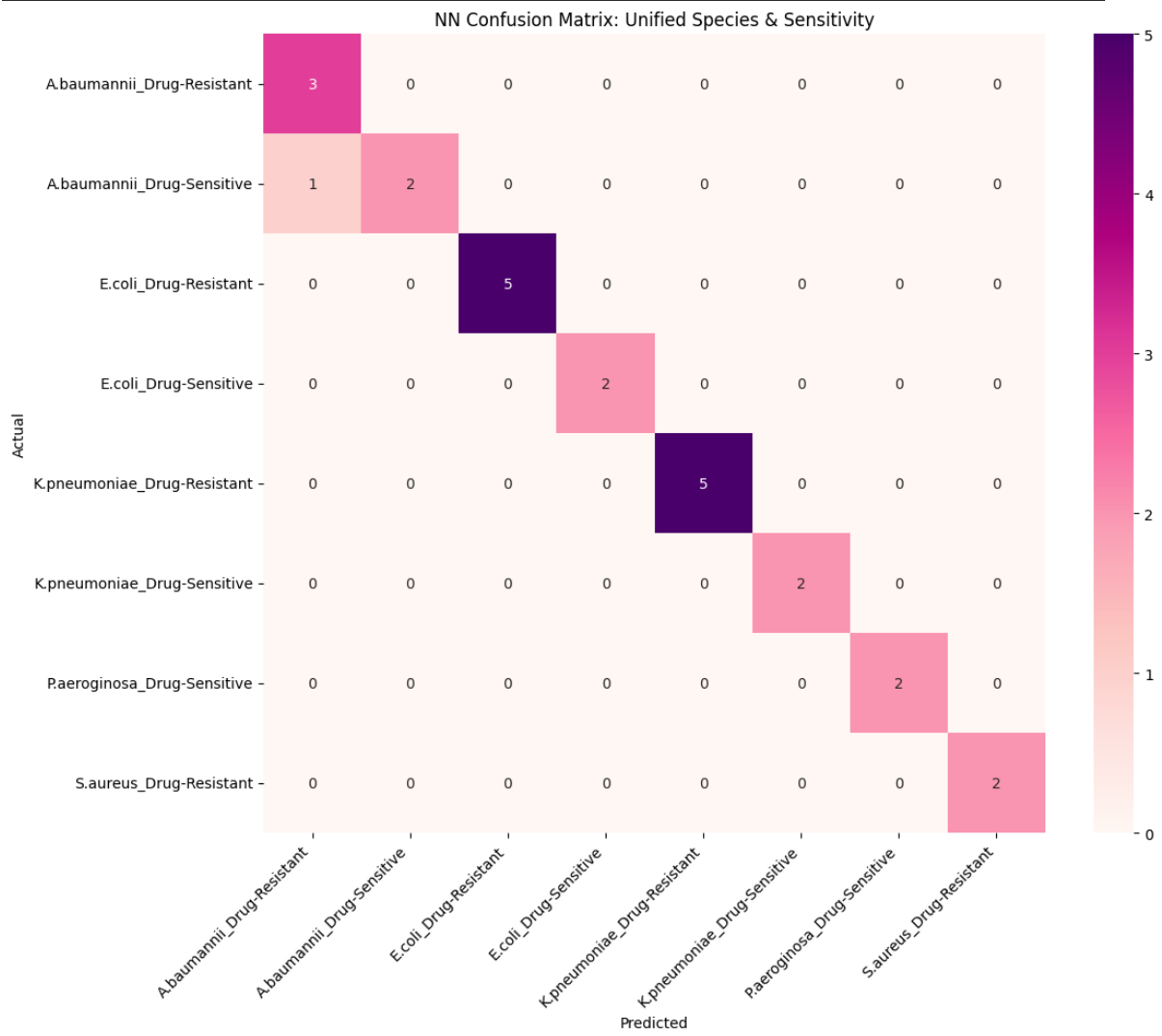
Unified Model Test Accuracy: 0.9583

1/1 0s 71ms/step

Unified Classification Report:

	precision	recall	f1-score	support
A.baumannii_Drug-Resistant	0.75	1.00	0.86	3
A.baumannii_Drug-Sensitive	1.00	0.67	0.80	3
E.coli_Drug-Resistant	1.00	1.00	1.00	5
E.coli_Drug-Sensitive	1.00	1.00	1.00	2
K.pneumoniae_Drug-Resistant	1.00	1.00	1.00	5
K.pneumoniae_Drug-Sensitive	1.00	1.00	1.00	2
P.aeruginosa_Drug-Sensitive	1.00	1.00	1.00	2
S.aureus_Drug-Resistant	1.00	1.00	1.00	2

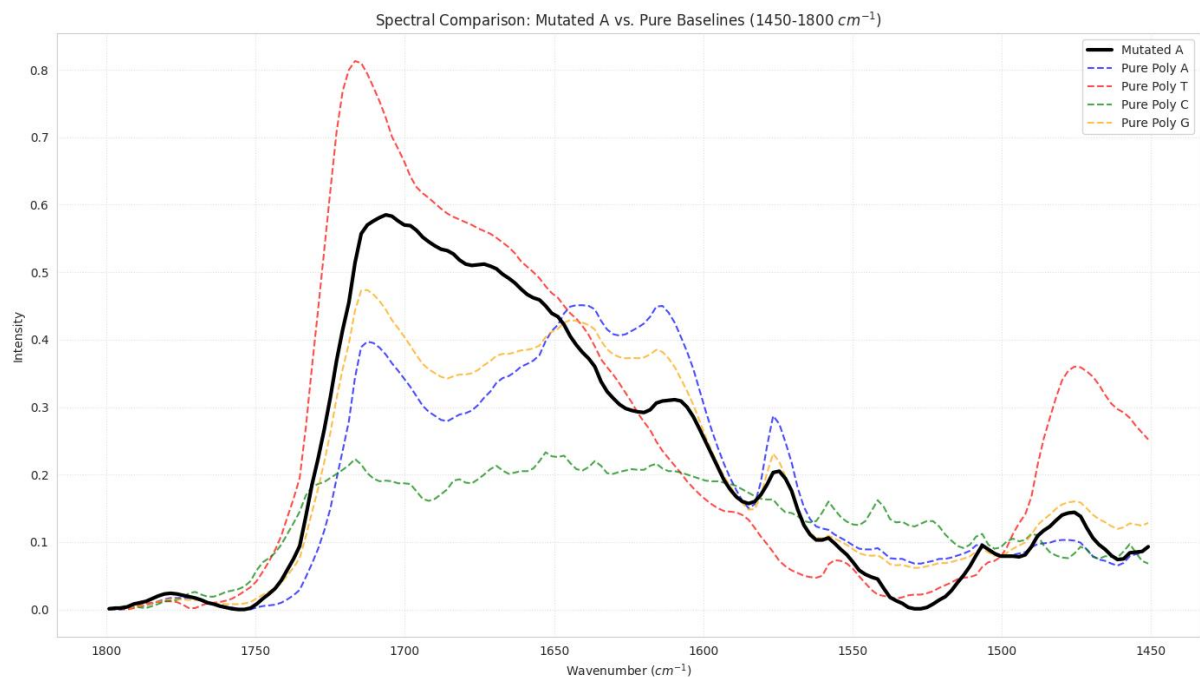
accuracy			0.96	24
macro avg	0.97	0.96	0.96	24
weighted avg	0.97	0.96	0.96	24



Nucleotide composition data

As we remember from the t-SNE tests of this data, there is some clusters at specific wavenumbers that have considerable variability, but overall the plot looked very complex for simple classification.

Since the mutated genes are composed of various proportions of pure nucleotides, naturally I wished to find the relationship between proportions and spectral intensity. Shroyon Das also had used a linear relationship when he was comparing mutated spectra against weighted averages. (L. Mariey, 2001)



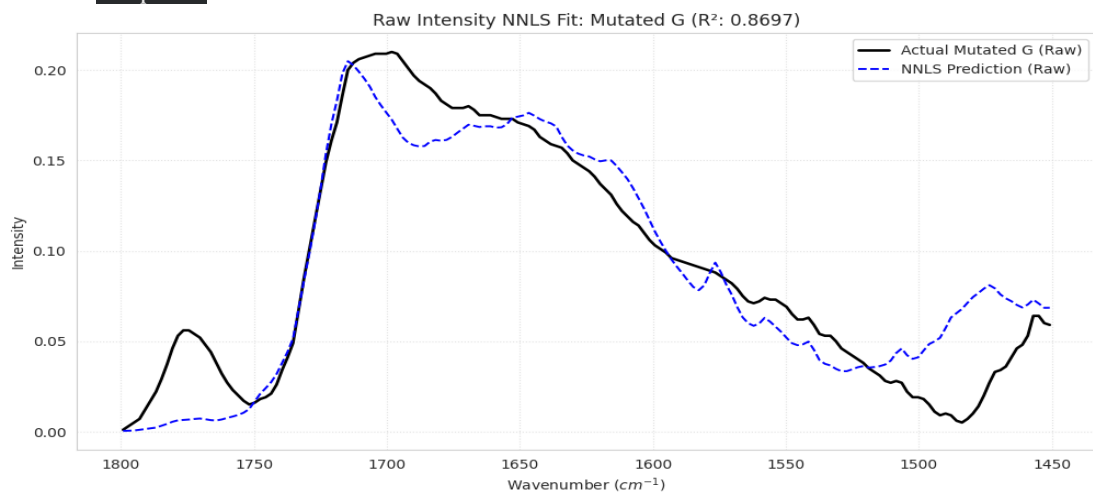
Let us say our composition was $aA+tT+gG+cC$ for a certain mutated gene. I had identified the wavenumbers that matter most for classification, and attributed the rest to noise.

Using that range as a target, where a predicted composition (since it is assumed a linear relationship) can produce a reconstructed plot that is the best optimised fit for the actual mutated plot. This [Optimization problem](#) can be attempted using a regression model.

Below is the algorithm used for Non Negative Least Squares Regression([NNLS](#)) (Bro, 1999)

- **Inputs:**
 - a real-valued matrix A of dimension $m \times n$,
 - a real-valued vector y of dimension m ,
 - a real value ϵ , the tolerance for the stopping criterion.
- **Initialize:**
 - Set $P = \emptyset$.
 - Set $R = \{1, \dots, n\}$.
 - Set x to an all-zero vector of dimension n .
 - Set $w = A^T(y - Ax)$.
 - Let w^R denote the sub-vector with indexes from R
- **Main loop:** while $R \neq \emptyset$ and $\max(w^R) > \epsilon$:
 - Let j in R be the index of $\max(w^R)$ in w .
 - Add j to P .

- Remove j from R .
- Let A^P be A restricted to the variables included in P .
- Let s be vector of same length as x .
- Let s^P denote the sub-vector with indexes from P .
- Let s^R denote the sub-vector with indexes from R .
- Set $s^P = ((A^P)^T A^P)^{-1} (A^P)^T y$
- Set s^R to zero
- While $\min(s^P) \leq 0$:
 - Let $\alpha = \min x_i/x_i - s_i$ for i in P where $s_i \leq 0$.
 - Set x to $x + \alpha(s - x)$.
 - Move to R all indices j in P such that $x_j \leq 0$.
 - Set $s^P = ((A^P)^T A^P)^{-1} (A^P)^T y$
 - Set s^R to zero.
- Set x to s .
- Set w to $A^T(y - Ax)$.
- Output: x



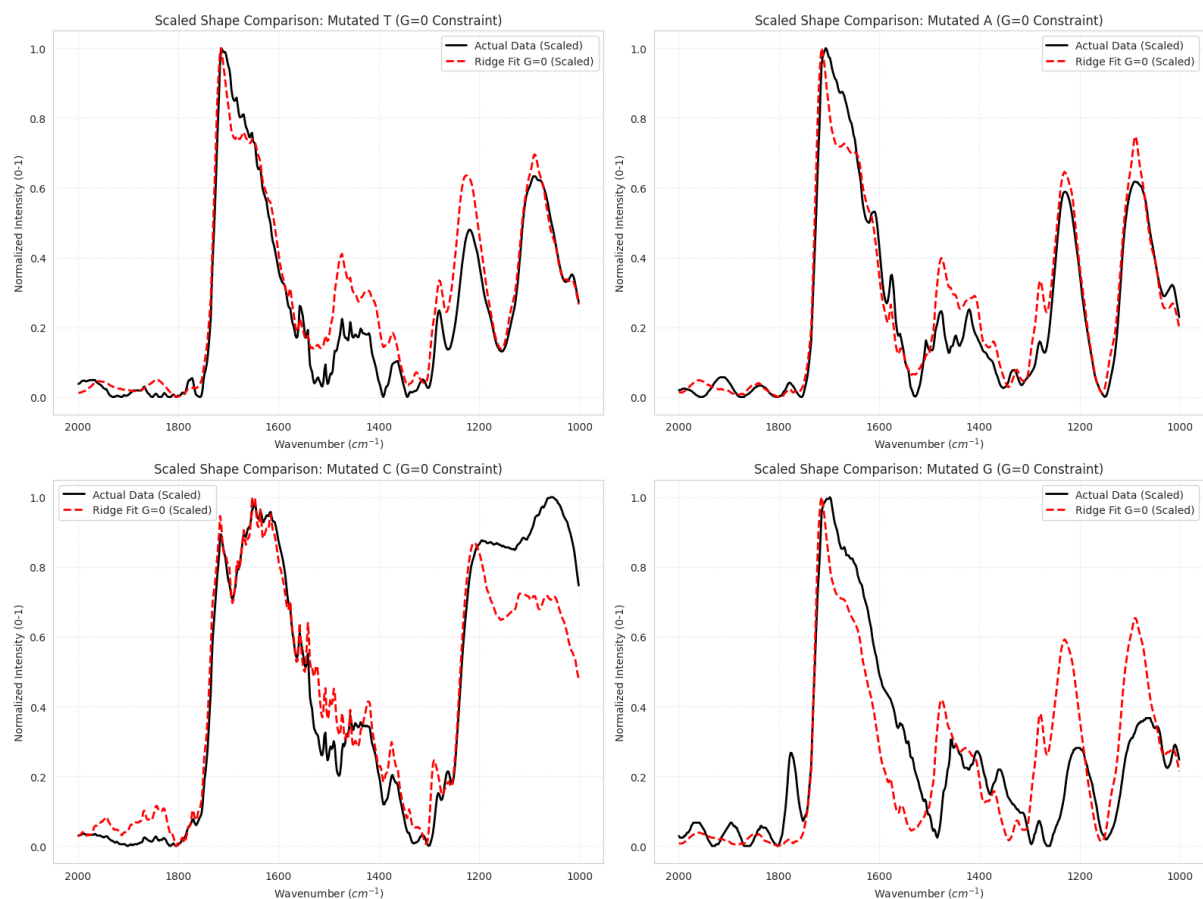
NNLS regression model fit like the above figure as it predicted the following raw intensity percentage. It has an R^2 score of 0.8697, where 1 is the perfect fit, so I wished to use models to improve on this metric but none of them could predict the actual composition

Mutated A has 8A,1C,3T.....*****

--- Comprehensive Composition Comparison: Mutated G ---

	Nucleotide	NNLS (Derivative) %	NNLS (Raw Intensity) %	MLP-PCA (Non- Linear) %
0	Poly A	15.29	32.27	30.07
1	Poly T	27.17	27.94	1.07
2	Poly C	0.00	8.76	40.71
3	Poly G	57.54	31.02	28.15

The models could not get the hang of it. Then I used the number of points overlapping between mutated samples and poly sample as a proxy for whether the sample has a certain nucleotide, where any overlap less than 2 percent normalised was considered as noise. But there was a huge, unexplained representation of poly G in the mutated species. So G was set to 0 as a starting point. The following are the comparisons of the reconstructed and the actual mutated samples.

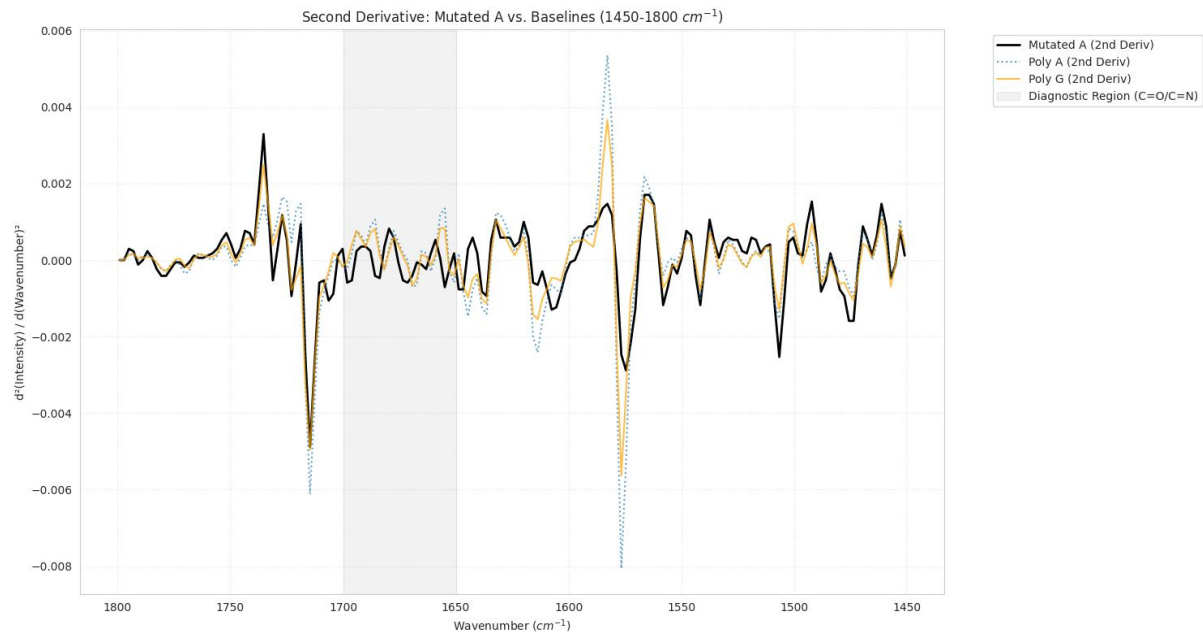


0We see a trend where it is a good fit in the region from 1750 to 1600 cm^{-1} , essentially traces the slope there, but struggles elsewhere. But mutated C has a really good fit for longer, and it could predict the composition for poly A and C quite accurately, as seen below.

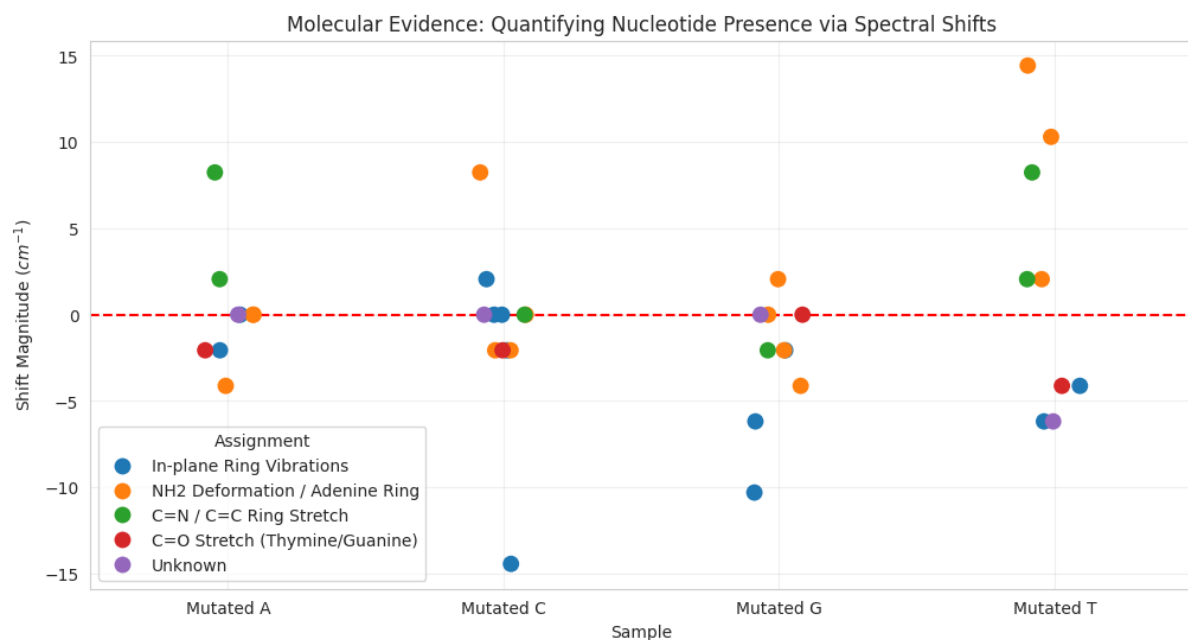
-- Cross-Sample Optimization Validation --

	Actual %				Predicted %			
Nucleotide	A	C	G	T	A	C	G	T
Sample								
Mutated A	69.23	7.69	0.00	23.08	24.02	11.54	26.32	38.12
Mutated C	8.33	66.67	25.00	0.00	8.50	60.63	0.00	30.86
Mutated G	8.33	25.00	66.67	0.00	0.00	10.59	89.41	0.00
Mutated T	25.00	0.00	8.33	66.67	24.70	19.67	15.47	40.16

I changed my method and checked where mutated genes were closer to the poly genes when we take on the second derivative data, as we saw some slope correlation before. In this plot we observe that there are some peaks that seem shifted across the wavenumber.



I used known vibrations corresponding to different wavenumber regions and plotted how much shift in wavenumber there is in those regions when mutated genes were compared with the mean spectra of all the poly samples. These vibrations are evidence for specific types of base pairs.



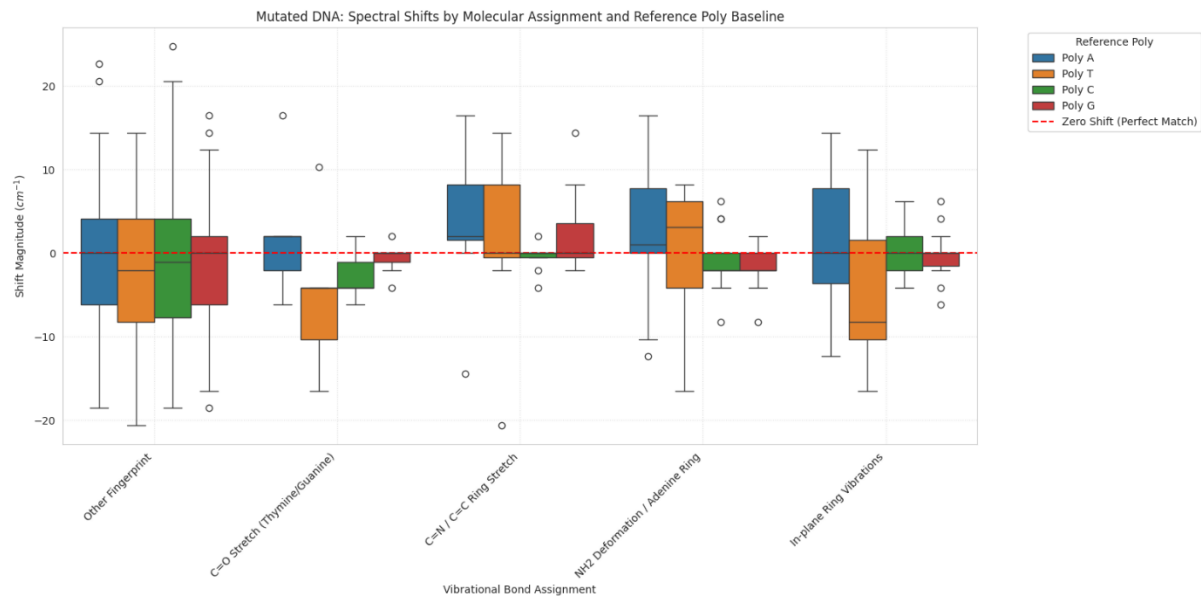
This gave me the following analysis of the mutated samples of their compositions.

[Mutated T]: Contains Thymine (Major), Adenine (Minor), Guanine (Trace)

[Mutated G]: Contains Guanine (Major), Cytosine (Minor), Adenine (Trace)

[Mutated C]: Contains Cytosine (Major), Guanine (Minor), Adenine (Trace)

[Mutated A]: Contains Adenine (Major), Thymine (Minor), Cytosine (Trace)

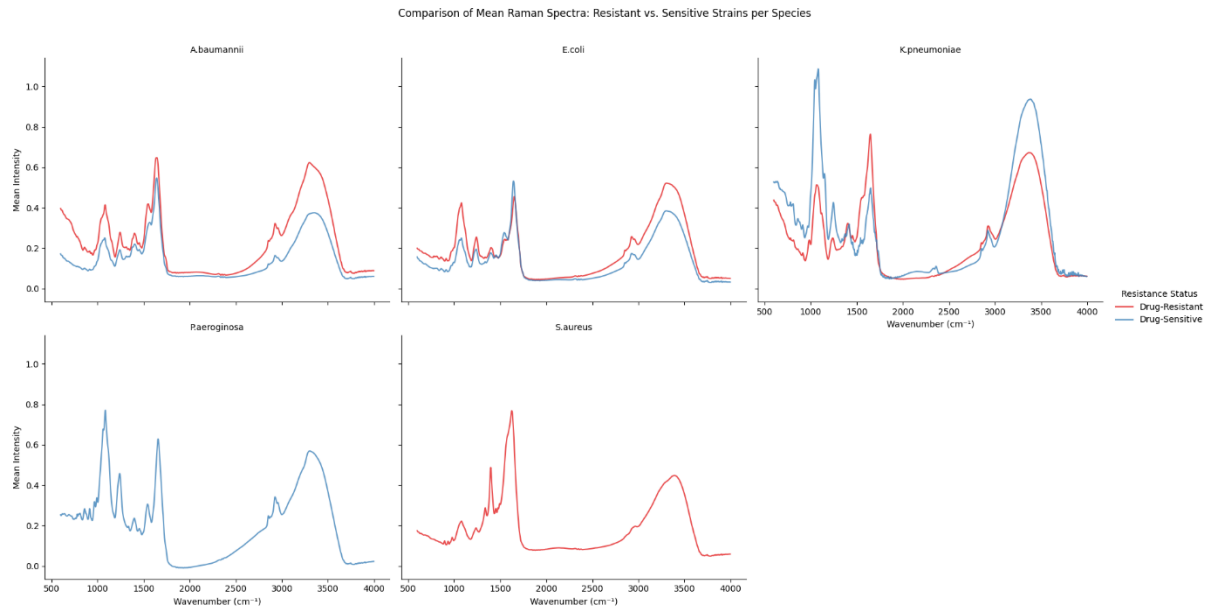


I also wanted to check the shift magnitude when poly samples are compared to their magnitude if it can actually be a reliable predictor of bases in the data.

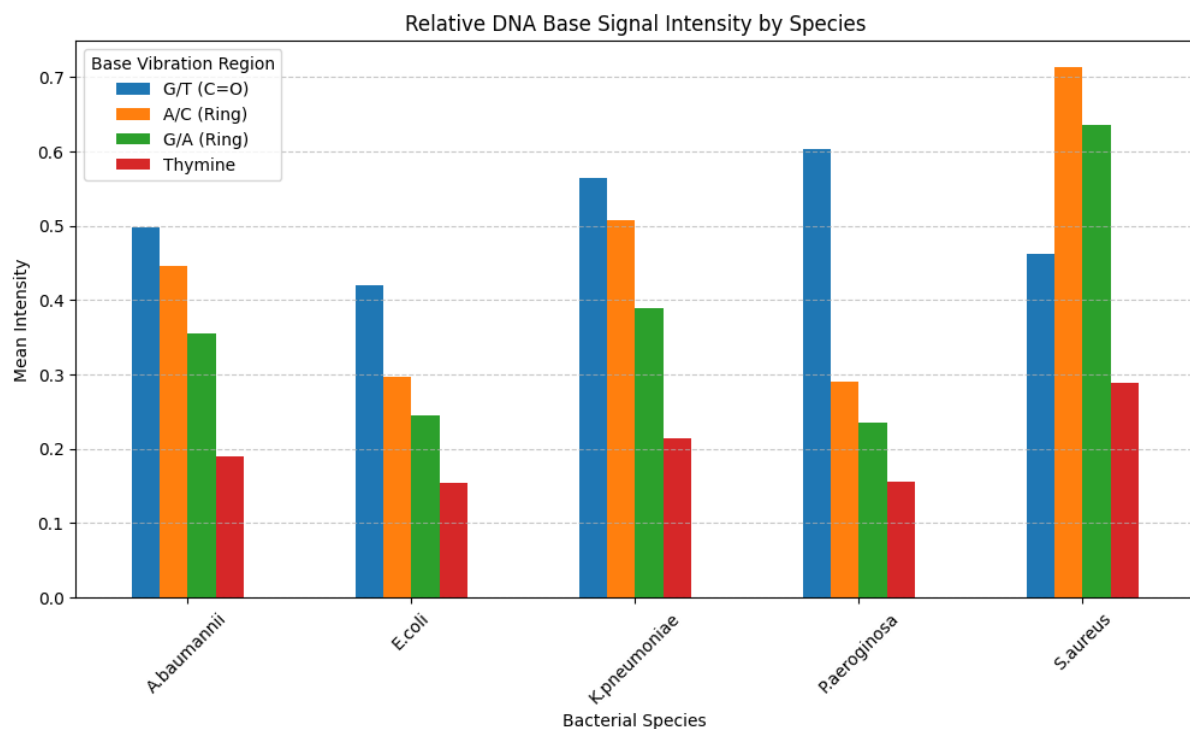
They appear to be somewhat reasonable predictors of bases but not their exact composition.

Results and Discussion

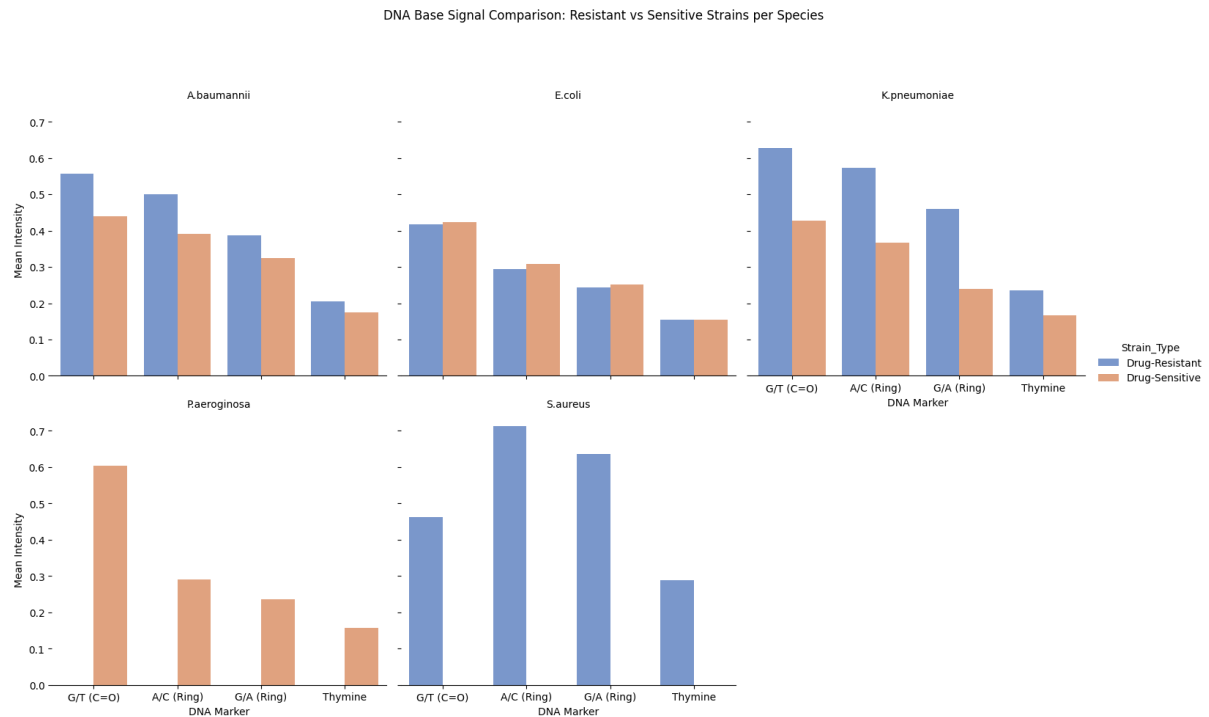
I could build a model that can predict the species and strain of the bacteria from just its FTIR spectral data. But my goal was to find the actual markers in the data that can be used directly for identification.



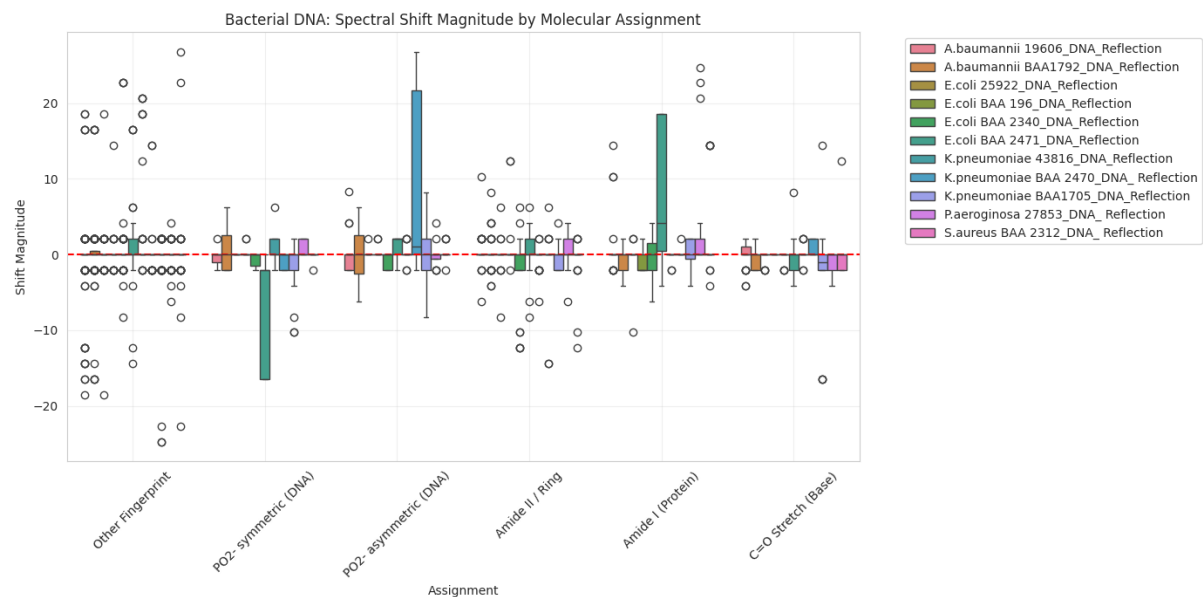
From the PCA components and from the work done on base pair data, we found out the wavenumbers to focus on for genetic mutation analysis.



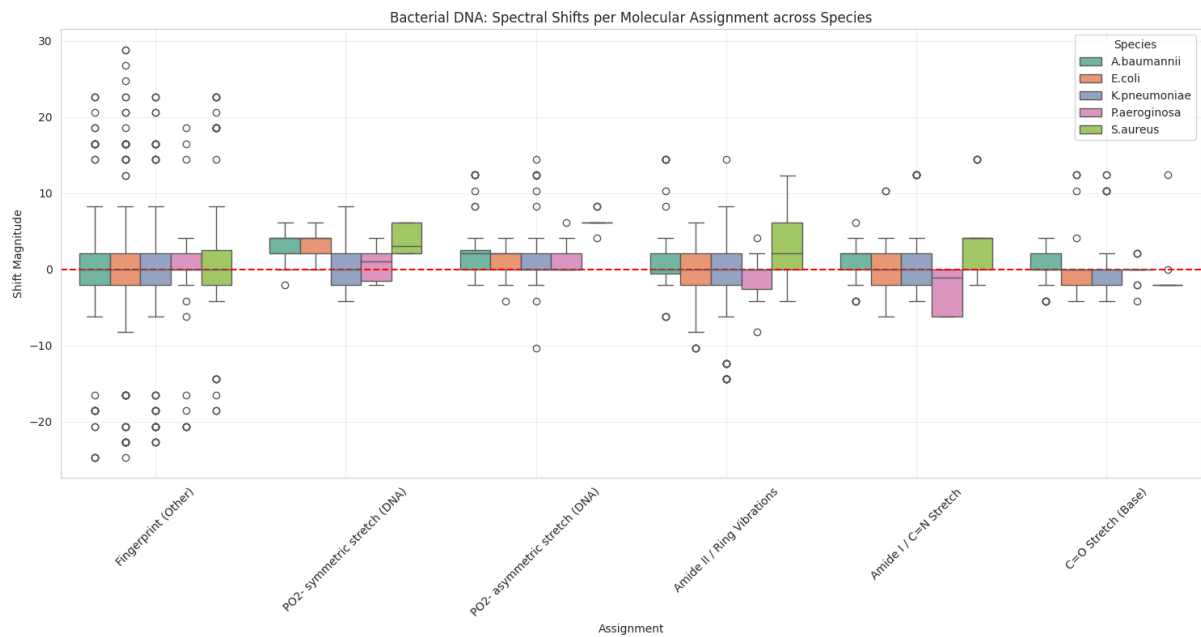
Each of these vibrations' intensity corresponds to one or two nucleotides. If evidence from two vibrations corroborates a nucleotide, we assume its strong presence.



Although not exactly giving us the composition, these vibrations have enough information to be able to classify from FTIR data.



Here, I used the shift magnitude while comparing bacterial DNA to the mean of all poly samples. The box plot above is for all the samples, and the one on the next page is for each species as a whole.



Conclusion and Future Scope

This report concludes that although extracting proportions out of FTIR is difficult, proxies like shift magnitude and PCA have enough information to classify the species and drug sensitivity of the Bacterial DNA samples.

I had faced issues given the complex nature of the Nucleotide Composition Data. I wish for this work can give a better idea later as to find a way to use FTIR for sensitive base pair identification.

A greater number of samples of these bacterial DNA could be obtained to validate these regression and neural network data, and a model that can identify a multitude of pathogens just from FTIR can be built.

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